

Paper IV — The Cross-Layer Evaluation

When the Attacker Is Indistinguishable from the Weather: A Cross-Layer Benchmark and a Unified Perturbation Budget for Certified UAV Autonomy

Research proposal from the CERTIFLIGHT concept (WP2/WP3) · RobustUAVs.ai · Roger Nick Anaedevha · 2026-10-01

Target: USENIX Security · extends the parent manuscript [1]

Abstract

A UAV’s navigation stack cannot tell, from the signal alone, whether its position error is caused by a jammer, a drifting inertial unit, or a gust. We take that indistinguishability as a security principle rather than a nuisance. If a certificate is to mean anything to an operator, it must bound the mission outcome under *whatever* produced the perturbation, because the attacker is free to choose the component that the defence models least well. We develop a unified perturbation budget $\delta_{\text{tot}} = \delta_{\text{net}}(\theta) \oplus \delta_{\text{gnss}} \oplus \delta_{\text{fault}} \oplus \delta_{\text{env}}$ that aggregates an adversarial residual (a function of a network detector’s operating point), GNSS denial, stochastic faults and bounded environmental forcing into one certified floor on the mission-completion rate — and we show the security consequence the single-source view misses: an adversary who shapes its interference to *mimic* a fault or a gust evades a detector tuned to malicious signatures while producing the same navigation degradation. The evaluation substrate is the missing ingredient in this field: a reproducible cross-layer benchmark whose unit of record is the pairing of a network or environmental event with the navigation window it induces, each pairing carrying an auditable provenance flag that states whether the coupling was measured on one platform, derived from a physical model, or aligned in simulation. We quantify what the union-bound aggregation costs in tightness, and name the conditions under which a joint treatment recovers it.

1 The security argument

Three communities each bound one layer and leave the interface to assumption. Network intrusion detection bounds what reaches the aircraft but says nothing about the flight after a missed intrusion. Certified robustness [2, 4] bounds what a model does with a perturbation but not where it comes from or how often an adversary can make one. Fault-tolerant control rejects disturbances but has not been composed with either. The property an operator, insurer and regulator actually need — will this mission complete, and can we prove it? — falls between them.

The security content is sharper than “let us unify three budgets”. It is an **adversarial-equivalence** observation:

From the navigation input alone, an adversary’s shaped interference, a sensor fault, and an environmental disturbance are indistinguishable. An attacker will therefore choose the disguise the defender models least well. A certificate that is sound only against a declared attack model is unsound against the attacker who dresses as the weather.

This is why the budget must be source-agnostic to be a security guarantee at all, and why a detector tuned to malicious signatures is the wrong last line: the decisive attack is the one that looks benign. A physical-invariant monitor such as SAVIOR [6] checks the dynamics rather than the signature and is the right comparison point; our contribution is to carry its spirit through to a *certified mission floor* and to a benchmark that can measure the evasion.

2 The unified budget

Navigation state $\dot{x} = F(x, u)$, F Lipschitz with constant L over the operating region, horizon T . A bounded input perturbation $\|\delta u\| \leq \delta_{\text{tot}}$ gives, through Grönwall, a trajectory-tube radius $\rho = \delta_{\text{tot}} e^{LT}$; the mission completes while the tube fits the safe-corridor margin m , so the certified floor is $f(\delta_{\text{tot}}) = \Pr[\delta_{\text{tot}} e^{LT} \leq m]$ (parent manuscript).

The new object is the budget itself:

$$\delta_{\text{tot}} = \delta_{\text{net}}(\theta) \oplus \delta_{\text{gnss}} \oplus \delta_{\text{fault}} \oplus \delta_{\text{env}}, \quad (1)$$

aggregating sources with *different probabilistic semantics*: $\delta_{\text{net}}(\theta)$ deterministic worst-case (adversarial residual below detector threshold θ , bounded by the residual-budget lemma $\Delta(\theta) \leq H \min(\theta, s)$), δ_{gnss} and δ_{fault} stochastic, δ_{env} bounded-but-unknown.

Proposition 1 (Heterogeneous composition, with its tightness cost). *The $\oplus$ of a deterministic, a stochastic and a bounded-unknown source admits a joint certificate. The union bound over independently certified sources is sound but loose; the looseness is exactly the dependence-uncertainty gap of Makarov’s bound [5, 3] on the distribution of a sum with fixed marginals and unknown coupling. A joint treatment recovers the union bound’s slack precisely when the sources are comonotone, and we characterise the intermediate regime.*

■ *Naming the tightness gap as the Makarov/VaR-aggregation gap is the theoretical hinge, and it is a claim to prove, not assert: the mapping from “three perturbation sources with mixed semantics” to “sum of random variables with fixed marginals and unknown copula” has to be made rigorous, and the deterministic source needs a degenerate-marginal treatment. If that reduction does not hold cleanly, the paper falls back to the plain union bound — which is already sound — and reports the gap empirically instead.*

3 The benchmark (the field’s missing substrate)

A guarantee spanning two layers needs data objects that record the coupling, and no public corpus does. The benchmark’s unit of record is the **cross-layer pairing**: a network or environmental window joined to the navigation window it induces, carrying the detector operating point, the residual budget, the induced perturbation, and the mission outcome.

Provenance is mandatory and enforced. Every pairing carries a flag: `measured_same_platform` (both windows observed on one airframe), `physics_model` (coupling through a stated physical relation), or `synthetic_alignment` (aligned in simulation). A reviewer filters by evidence strength rather than trusting an unaudited join. This is the discipline that keeps the central honest qualification visible: of six corpora, only one observes a network event and the flight it degrades on the same platform, and its release carries no machine-readable attack intervals — so the strongest evidence class is sparse, and the benchmark says so in every record rather than burying it.

Contents. Lossless ingestion adapters for six public corpora; a physics-informed navigation benchmark; a detector-operating-curve harness that sweeps θ and reports recall and false-positive rate (the network-side premise the composed guarantee consumes); and a reference pairing pipeline. Third-party data is fetched from origin under its own licence; adapters and unified labels are released, not redistributed corpora.

4 The evasion study (the USENIX core)

The benchmark makes the adversarial-equivalence claim measurable. The study:

RQ1 Can an attacker mimic a fault/gust to evade a signature-tuned detector while producing the same δ ? Construct shaped interference whose network-layer signature falls below θ but whose induced navigation degradation equals that of a modelled fault, and measure the detector’s miss rate against the induced δ .

RQ2 Does the composed certificate bound the mission outcome even when the detector is evaded? This is the payoff: the floor should hold because the budget is source-agnostic, where a detector-only defence fails silently.

RQ3 How loose is the union bound in practice, across the six corpora and the physics benchmark, and where does the joint treatment of Proposition 1 recover it?

RQ4 Soundness: does the certified floor lower-bound the measured completion rate at *every* operating point, across seeds, under Holm-corrected tests? A violation is a bug in the instantiation.

Why USENIX Security. The contribution is a threat-model correction (the attacker who dresses as the weather), an evaluation methodology with enforced provenance, and a released artifact — the shape of contribution USENIX rewards, and the venue where an evaluation substrate can become a community default. The theorem is the frame; the benchmark and the evasion measurement are the paper.

5 What is in hand, what is new

In hand (parent + Papers I–II)	New here
Six-corpus ingest, certificate engine, four certificate methods	δ_{fault} and δ_{env} as budget terms with mixed semantics
Residual-budget lemma; $\gamma = 1.625$ m/s campaign supremum	the adversarial-equivalence threat model and the evasion study
Detector-operating-curve harness; provenance flags	Proposition 1 and the Makarov/VaR tightness characterisation
HCRL UAVCAN, WHELAN and four more adapters	public benchmark release whose unit is the cross-layer pairing

6 Relation to the other papers

- **Parent manuscript** proves the composition for the time-delay class on the network side. Paper IV broadens the *source* of the perturbation to faults and environment and makes the threat model adversarial-equivalence; it is the security-and-measurement paper to the parent’s theory paper.
- **Paper I** varies γ and Δ as design variables; Paper IV varies the *source* of δ . Complementary axes of the same budget.
- **Paper III** consumes the per-agent radius this budget produces. Paper IV is upstream of it.
- **Paper II (SoK)** supplies the vocabulary; Paper IV is a $\mathbf{Q}_{op}$ -level security evaluation within it.

7 Risks

- **The equivalence attack may be only partially realisable.** Shaping interference to match a fault’s δ and stay below θ may not be achievable for every fault profile. That is itself a finding — the subset of degradations an attacker can disguise is the real attack surface — and the paper reports the achievable region rather than claiming the whole.
- **The tightness reduction (Prop. 1) may not close.** Fallback: the sound union bound plus an empirical gap measurement. The benchmark and evasion study stand regardless.
- **Same-platform pairings stay scarce.** The provenance flags make this explicit rather than hidden; `physics_model` and `synthetic_alignment` pairings sustain coverage, and the limitation is reported as a limitation. This is a standing constraint from the parent work, carried forward honestly.
- **Fault/gust realism.** δ_{fault} and δ_{env} are modelled from the fault-tolerant-control literature; every modelled number carries its provenance tag, and the paper does not present a simulated gust as a measured one.

8 Page plan and timeline

Section	pp	Section	pp
Introduction	1.0	Evasion study and results	2.5
Threat model (equivalence)	1.25	Soundness and union-bound gap	1.5
Unified budget and theory	1.75	Discussion, ethics, limits	1.0
Benchmark design	1.75	Related work, conclusion	1.0
Body (USENIX 13-page limit)			11.75

About five to six months. The ingest, harness and certificate engine exist; the new work is the fault and environment budget terms, the evasion construction, and the benchmark packaging and release. An ethics subsection is mandatory: the evasion method is dual-use, and following the parent’s evasion analysis, it is presented with its mitigation (invariant-based monitoring, and source-agnostic certification) in the same section, bounds what the attacker achieves, and targets a defensive artifact.

References

- [1] Anonymous. From Wire to Flight: A Conditional Mission-Completion Guarantee for UAVs. under submission, 2026. the parent manuscript; anonymised for review.
- [2] Jeremy M. Cohen, Elan Rosenfeld, and J. Zico Kolter. Certified Adversarial Robustness via Randomized Smoothing. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2019.
- [3] Paul Embrechts, Giovanni Puccetti, and Ludger Rüschendorf. Model Uncertainty and VaR Aggregation. *Journal of Banking and Finance*, 2013.
- [4] Thomas H. Grönwall. Note on the Derivatives with Respect to a Parameter of the Solutions of a System of Differential Equations. *Annals of Mathematics*, 20(4):292–296, 1919.
- [5] G. D. Makarov. Estimates for the Distribution Function of a Sum of Two Random Variables when the Marginal Distributions Are Fixed. *Theory of Probability and Its Applications*, 1981.
- [6] Raul Quinonez, Jairo Giraldo, Luis Salazar, Erick Bauman, Alvaro Cardenas, and Zhiqiang Lin. SAVIOR: Securing Autonomous Vehicles with Robust Physical Invariants. In *Proc. USENIX Security Symposium*, 2020.